

Polyuria causes — AVP deficiency vs AVP resistance vs polydipsia



1. AVP deficiency wing (central DI)

WHAT YOU SEE

Left wing labeled AVP deficiency, formerly central DI

WHAT IT ENCODES

Central DI (now called AVP deficiency) reflects failure of hypothalamic AVP synthesis or release. Copeptin is low and DDAVP corrects the urine. Causes cluster as trauma/surgery, tumor, infiltrative, and idiopathic.

BOARD TRAP

New nomenclature: central DI = AVP deficiency. Low/absent copeptin with DDAVP response confirms it versus nephrogenic disease.

2. Trauma / pituitary surgery

WHAT YOU SEE

Trauma surgery portrait, central DI cause

WHAT IT ENCODES

Head trauma and transsphenoidal/pituitary surgery are common causes of central DI, often via stalk damage. A classic triphasic response (DI, then SIADH, then permanent DI) can follow. Onset is frequently abrupt.

BOARD TRAP

Post-pituitary-surgery triphasic pattern: transient DI then a SIADH phase — overtreating the first phase with fluids worsens the later hyponatremia.

3. Tumor / mass lesion

WHAT YOU SEE

Tumor portrait in deficiency wing

WHAT IT ENCODES

Craniopharyngioma, germinoma, and metastases (lung, breast) can destroy hypothalamic AVP neurons. Pituitary adenomas rarely cause DI unless large; suprasellar lesions are more typical. Imaging the hypothalamic-pituitary axis is essential.

BOARD TRAP

A pituitary microadenoma alone rarely causes DI; new DI should prompt search for a suprasellar/hypothalamic process or infiltration.

4. Infiltrative — sarcoid/LCH

WHAT YOU SEE

Infectious/infiltrative portrait, lower left

WHAT IT ENCODES

Granulomatous and infiltrative disease (sarcoidosis, Langerhans cell histiocytosis) and lymphocytic hypophysitis target the stalk and hypothalamus. They cause stalk thickening and loss of the bright spot. Often accompanied by anterior pituitary deficits.

BOARD TRAP

Stalk thickening plus central DI suggests sarcoid, LCH, or hypophysitis — biopsy/systemic workup rather than assuming idiopathic DI.

5. Idiopathic / autoimmune

WHAT YOU SEE

Portrait labeled idiopathic, deficiency wing

WHAT IT ENCODES

A large fraction of central DI is idiopathic, many of which are autoimmune against AVP neurons. This is a diagnosis of exclusion after imaging and infiltrative workup. May later reveal an occult germinoma, so follow-up imaging is advised.

BOARD TRAP

Idiopathic central DI is a diagnosis of exclusion — repeat MRI is needed because an occult germinoma can declare itself later.

6. AVP resistance wing (nephrogenic)

WHAT YOU SEE

Center wing, AVP resistance, formerly nephrogenic DI

WHAT IT ENCODES

Nephrogenic DI (AVP resistance) is renal failure to respond to AVP. Copeptin is high and DDAVP does NOT concentrate the urine. Acquired causes far outnumber congenital ones.

BOARD TRAP

Nephrogenic DI has HIGH copeptin and no DDAVP response — the discriminator from central DI.

7. Lithium — most common cause

WHAT YOU SEE

Lithium bottle, most common acquired cause

WHAT IT ENCODES

Lithium is the most common acquired cause of nephrogenic DI, entering principal cells via ENaC and downregulating AQP2. The defect can persist after stopping lithium. Amiloride blocks ENaC entry and is protective.

BOARD TRAP

Lithium NDI: amiloride (blocks ENaC) is the targeted therapy; the defect may be irreversible even after discontinuation.

8. Hypercalcemia / hypokalemia

WHAT YOU SEE

Metabolic cause panel, AVP resistance wing

WHAT IT ENCODES

Hypercalcemia and hypokalemia impair medullary concentrating ability and AQP2 function, causing reversible nephrogenic DI. Correcting the electrolyte abnormality restores concentration. These are common, treatable metabolic causes.

BOARD TRAP

Polyuria with hypercalcemia is reversible NDI — treat the calcium rather than chasing a pituitary lesion.

9. Hereditary — V2 / AQP2

WHAT YOU SEE

Hereditary portraits, V2 receptor and AQP2

WHAT IT ENCODES

Congenital nephrogenic DI is most often X-linked from V2 receptor mutations (about 90%), with autosomal AQP2 mutations accounting for the rest. Presents in infancy with severe polyuria and hypernatremia. DDAVP fails to respond.

BOARD TRAP

Congenital NDI is usually X-linked V2 receptor (boys) — not AQP2 — and unlike central DI shows no DDAVP response.

10. Sickle cell nephropathy

WHAT YOU SEE

Sickle cell panel in resistance wing

WHAT IT ENCODES

Sickle cell disease/trait damages the renal medulla and vasa recta, producing a concentrating defect and nephrogenic DI. Hyposthenuria is an early finding. Reflects medullary microinfarction.

BOARD TRAP

Sickle cell hyposthenuria is a renal medullary concentrating defect (NDI physiology), not a pituitary problem.

11. Primary polydipsia wing

WHAT YOU SEE

Right wing labeled primary polydipsia

WHAT IT ENCODES

Primary polydipsia is excess water intake suppressing AVP appropriately, with dilute plasma and low-normal sodium. Distinguished by low/normal plasma osmolality and a high copeptin response to hypertonic saline. Includes psychogenic and dipsogenic forms.

BOARD TRAP

Primary polydipsia has LOW plasma sodium/osmolality (dilution), whereas DI tends toward HIGH-normal sodium from water loss.

12. Psychogenic polydipsia

WHAT YOU SEE

Portrait, psychogenic, polydipsia wing

WHAT IT ENCODES

Psychogenic polydipsia is compulsive water drinking, often in psychiatric patients. Chronic high intake can wash out the medullary gradient, blunting maximal concentration. Acute water intoxication can cause hyponatremia.

BOARD TRAP

In long-standing polydipsia the medullary gradient is washed out, so the water deprivation test can falsely mimic partial nephrogenic DI.

13. Dipsogenic lesion

WHAT YOU SEE

Dipsogenic lesion portrait, polydipsia wing

WHAT IT ENCODES

Dipsogenic DI is a downward reset of the thirst threshold, driving excessive drinking despite normal osmolality. AVP secretion is appropriately suppressed. DDAVP is risky here because continued drinking causes hyponatremia.

BOARD TRAP

Giving DDAVP to a dipsogenic polydipsic patient who keeps drinking precipitates dangerous dilutional hyponatremia.

14. Diagnostic lab first

WHAT YOU SEE

Banner: polyuria/polydipsia causes hall, see diagnostic lab first

WHAT IT ENCODES

Workup starts by confirming true polyuria (>3 L/day or >50 mL/kg/day) and excluding osmotic diuresis like hyperglycemia. Then plasma sodium/osmolality, urine osmolality, and copeptin-based testing localize the lesion. Sequence is urine then sodium then copeptin then MRI.

BOARD TRAP

Always exclude osmotic diuresis (glucose, mannitol, high urea) before labeling polyuria as DI — these cause polyuria with normal AVP axis.
